Actividad 2 - Raíces de ecuaciones

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PRIMER PUNTO

$\boldsymbol{a}.\ \ \boldsymbol{x = \frac{1}{(x+1)^2}}$

$$x_1 = \frac{1}{(1+1)^2} = 0.25$$
$$x_2 = \frac{1}{(0.25+1)^2} = 0.64$$
$$x_3 = \frac{1}{(0.64+1)^2} = 0.37$$
$$x_4 = \frac{1}{(0.37+1)^2} = 0.53$$
$$x_5 = \frac{1}{(0.53+1)^2} = 0.42$$

$\boldsymbol{b}.\ \ \boldsymbol{x = \left(\frac{6 - x - x^3}{4}\right)^{\frac{1}{2}}}$

$$x_1 = \left(\frac{6-1-1^3}{4}\right)^2 = \left(\frac{6-1-1}{4}\right)^{\frac{1}{2}} = \left(\frac{4}{4}\right)^{\frac{1}{2}} = 1$$
$$x_2 = \left(\frac{6-1-1^3}{4}\right)^2 = \left(\frac{6-1-1}{4}\right)^{\frac{1}{2}} = \left(\frac{4}{4}\right)^{\frac{1}{2}} = 1$$
$$x_3 = \left(\frac{6-1-1^3}{4}\right)^2 = \left(\frac{6-1-1}{4}\right)^{\frac{1}{2}} = \left(\frac{4}{4}\right)^{\frac{1}{2}} = 1$$

$\boldsymbol{c}.\ \ \boldsymbol{x = \sin(x)}$

$$x_1 = \sin(0.5) = 0.4794$$
$$x_2 = \sin(0.5) = 0.4615$$
$$x_3 = \sin(0.5) = 0.4455$$
$$x_4 = \sin(0.5) = 0.4314$$
$$x_5 = \sin(0.5) = 0.4181$$

$\boldsymbol{d}.\ \boldsymbol{\tan tanx = \ln lnx}$

$$g(x) = tan^{-1}\left(tan^{-1}(\ln(\ln(x)))\right)$$
$$x_1 = tan^{-1}\left(tan^{-1}\left(ln\ (\ln(2))\right)\right)$$
$$tan^{-1}\left(tan^{-1}\left(ln\ (0.693)\right)\right)$$
$$tan^{-1}\left(tan^{-1}(-0.366)\right)$$
$$tan^{-1}(-0.350)$$

-0.339

$\boldsymbol{e.}\ \ \boldsymbol{x = 4 + (\frac{x-1}{x+1})}$

$x_1 = 4 + (\frac{2-1}{2+1}) = 4 + (\frac{1}{3}) = 4 + 0.333 = 4.333$

$x_2 = 4 + (\frac{4.333-1}{4.333+1}) = 4 + (\frac{3.333}{5.333}) = 4 + 0.625 = 4.625$

$x_3 = 4 + (\frac{4.625-1}{4.625+1}) = 4 + (\frac{3.625}{5.625}) = 4 + 0.644 = 4.644$

$x_4 = 4 + (\frac{4.644-1}{4.664+1}) = 4 + (\frac{3.644}{5.644}) = 4 + 0.646 = 4.646$

$x_5 = 4 + (\frac{4.646-1}{4.646+1}) = 4 + (\frac{3.646}{5.646}) = 4 + 0.646 = 4.646$

$\boldsymbol{f.}\ \ \boldsymbol{x = \frac{secsec\ x}{2}}$

$x_1 = \frac{\sec(\sec(1))}{2} = \frac{\sec(1.557)}{2} = \frac{6.586}{2} = 3.293$

$x_2 = \frac{\sec(\sec(3.293))}{2} = \frac{\sec(1.022)}{2} = \frac{1.539}{2} = 0.769$

$x_3 = \frac{\sec(\sec(0.769))}{2} = \frac{\sec(1.415)}{2} = \frac{7.308}{2} = 3.654$

$x_4 = \frac{\sec(\sec(3.654))}{2} = \frac{\sec(1.065)}{2} = \frac{1.594}{2} = 0.797$

$x_5 = \frac{\sec(\sec(0.797))}{2} = \frac{\sec(1.429)}{2} = \frac{7.479}{2} = 3.739$

SEGUNDO PUNTO

$x^3 - 2x - 2 = 0$

1. $\boldsymbol{g(x) = \sqrt[3]{2x + x}}$

$$g'(x) = \frac{d}{dx}\left((2x+2)^{\frac{1}{3}} \right) = \frac{2}{3}(2x+2)^{-2/3}$$

2. $\boldsymbol{g(x) = \sqrt{2 + \frac{2}{x}}}$

$$g'(x) = \frac{d}{dx}\left(\left(2+\frac{2}{x}\right)^{\frac{1}{3}}\right) = \frac{-1}{x^2} * \frac{1}{\sqrt{2+\frac{2}{x}}}$$

3. $\boldsymbol{g(x) = x = \frac{2}{x} + \frac{2}{x^2}}$

$$g'(x) = -\frac{2}{x^2} - \frac{4}{x^3}$$

$\boldsymbol{a}$) $x = \sqrt[3]{2x + x}$

$$g'(1.5) = \frac{2}{3}\,(2(1.5)+2)^{-\frac{2}{3}} = \frac{2}{3}(5)^{-2/3} = 0.35$$

$\boldsymbol{b}$) $x = \sqrt{2 + \frac{2}{x}}$

$$g'(1.5) = \frac{-1}{(1.5)^2} * \frac{1}{\sqrt{2+\frac{2}{1.5}}} = -0.15$$

$\boldsymbol{c}$) $x = \frac{2}{x} + \frac{2}{x^2}$

$$g'(1.5) = -\frac{2}{(1.5)^2} - \frac{4}{(1.5)^3} = -0.96$$

CONCLUSION

El valor de *g'(x)* mas pequeño en magnitud corresponde a *g'(x)* $\sqrt{2+\frac{2}{x}}$ esto sugiere que dicha función tiene mejor potencial de convergencia en el intervalo (1,2)

1. $\boldsymbol{g(x) = \sqrt[3]{2x+2}}$

$x_1 = \sqrt[3]{2(1.5)+2} = 1.710$

$x_2 = \sqrt[3]{2(1.710)+2} = 1.771$

$x_3 = \sqrt[3]{2(1.771)+2} = 1.793$

2. $\boldsymbol{g(x) = \sqrt{2 + \frac{2}{x}}}$

$$x_1 = \sqrt{2 + \frac{2}{1.5}} = 1.632$$

$$x_2 = \sqrt{2 + \frac{2}{1.632}} = 1.598$$

$$x_1 = \sqrt{2 + \frac{2}{1.598}} = 1.584$$

$$\mathbf{3.}\ \boldsymbol{g(x) = x = \frac{2}{x} + \frac{2}{x^2}}$$

$$x_1 = \frac{2}{1.5} + \frac{2}{(1.5)^2} = 1.644$$

$$x_2 = \frac{2}{1.644} + \frac{2}{(1.644)^2} = 1.592$$

$$x_3 = \frac{2}{1.594} + \frac{2}{(1.594)^2} = 1.572$$

COMPARACIONES

- El método $g(x) = \sqrt{2 + \frac{2}{x}}$ converge de manera más rápida y estable en el intervalo (1,2), lo cual permite encontrar la raíz en ese intervalo con mas facilidad.

TERCER PUNTO

$$\boldsymbol{x_n + 1 = x_n - \frac{f(x_n)}{f'(x_n)}}$$

$$\boldsymbol{a)\ xe^x - 2 = 0}$$

$$f(x_0) = 1 * e^1 - 2 = e - 2 = 2.718 - 2 = 0.718$$

$$f'(x_0) = e^1 + 1 * e^1 = 2e = 5.436$$

$$x_1 = 1 - \frac{0.718}{5.436} = 1 - 0.1321 = 0.8679$$

$$f(x_1) = 0.8679 * e^{0.8679} - 2 = 0.8679 * 2.382 - 2 = 2.065 - 2 = 0.065$$

$$f'(x_1) = e^{0.8679} + 0.8679 * e^{0.8679} = 2.382 + 0.8679 * 2.382 = 2.382 + 2.066 = 4.448$$

$$x_2 = 0.8679 - \frac{0.065}{4.448} = 0.8679 - 0.0146 = 0.8533$$

$$f(x_2) = 0.8533 * e^{0.8533} - 2 = 0.8533 * 2.347 - 2 = 2.002 - 2 = 0.002$$

$f'^{(x_2)} = 2.347 + 0.8533 * 2.347 = 4.345$

$$x_3 = 0.8533 - \frac{0.002}{4.345} = 0.8533 - 0.00046 = 0.852$$

$\boldsymbol{b)\ x - 2coscosx = 0}$

$f'(x) = 1 + 2\sin(\cos(x)) * \sin(x)$

$f(x_0) = 1 - 2\cos(\cos(1)) = 1 - 2\cos(0.5403) = 1 - 2 * 0.8576 = 1 - 1.7152 = -0.7152$

$f'(x_0) = 1 + 2\sin(\cos(1)) * \sin(1) = 1 + 2 * 0.5144 * 0.8415 = 1 + 0.8659 = 1.8659$

$$x_1 = 1 - \frac{-0.7152}{1.8659} = 1 + 0.3834 = 1.3834$$

$f(x_1) = 1.3834 - 2\cos(\cos(1.3834)) = 1.3834 - 2\cos(0.1872) = 1.3834 - 2 * 0.9825$
$= 1.3834 - 1.965 = -0.5816$

$f'(x_1) = 1 + 2\sin(\cos(1.3834)) * \sin(1.3834) = 1 + 2 * 0.1861 * 0.9825 = 1 + 0.3654$
$= 1.3654$

$$x_2 = 1.3834 - \frac{-0.5816}{1.3654} = 1.3834 + 0.4258 = 1.8092$$

$f(x_2) = 1.8092 - 2\cos(\cos(1.8092)) = 1.8092 - 2\cos(-0.2359) = 1.8092 - 2 * 0.9723$
$= 1.8092 - 1.9446 = -0.1354$

$f'(x_2) = 1 + 2\sin(\cos(1.8092)) * \sin(1.8092) = 1 + 2 * -0.2343 * 0.9716 = 1 + (-0.4554)$
$= 0.5446$

$$x_3 = 1.8092 - \frac{-0.1354}{0.5446} = 1.8092 + 0.2487 = 2.0579$$

$\boldsymbol{c)\ lnlnx - x + 2 = 0}$

$$f'(x) = \frac{1}{xln(x)} - 1$$

$f(x_0) = \ln(\ln(5)) - 5 + 2 = \ln(1.6094) - 5 + 2 = 0.4759 - 5 + 2 = -2.5241$

$$f'^{(x_0)} = \frac{1}{5\ln(5)} - 1 = \frac{1}{5 * 1.6094} - 1 = 0.1242 - 1 = -0.8758$$

$$x_1 = 5 - \frac{-2.5241}{-0.8758} = 5 - 2.8822 = 2.1178$$

$f(x_1) = \ln(\ln(2.1178)) - 2.1178 + 2 = \ln(0.7498) - 2.1178 + 2 = -0.2886 - 2.1178 + 2$
$= -0.4064$

$$f'(x_1) = \frac{1}{2.1178\ln(2.1178)} - 1 = \frac{1}{2.1178 * 0.7498} - 1 = 0.6292 - 1 = -0.3708$$

$$x_2 = 2.1178 - \frac{-0.4064}{-0.8758} = 2.1178 - 1.096 = 1.0218$$

$$f(x_2) = \ln(\ln(1.0218)) - 1.0218 + 2 = \ln(0.0215) - 1.0218 + 2 = -3.835 - 1.0218 + 2 = -2.8568$$

$$f'(x_2) = \frac{1}{1.0218 * \ln(1.0218)} - 1 = \frac{1}{1.0218 * 0.0215} - 1 = 45.46 - 1 = 44.46$$

$$x_3 = 1.0218 - \frac{-2.8568}{44.46} = 1.0218 + 0.0643 = 1.0861$$

$\boldsymbol{d)\ x^3 - 5x = -1}$

$$f'(x) = 3x^2 - 5$$

$$f(x_0) = (1)^3 - 5(1) + 1 = 1 - 5 + 1 = -3$$

$$f'(x_0) = 3(1)^2 - 5 = 3 - 5 = -2$$

$$x_1 = 1 - \frac{-3}{-2} = 1 + 1.5 = 2.5$$

$$f(x_1) = (2.5)^3 - 5(2.5) + 1 = 15.625 - 12.5 + 1 = 4.125$$

$$f'(x_1) = 3(2.5)^2 - 5 = 3(6.25) - 5 = 18.75 - 5 = 13.75$$

$$x_2 = 2.5 - \frac{4.125}{13.75} = 2.5 - 0.3 = 2.2$$

$$f(x_2) = (2.2)^3 - 5(2.2) + 1 = 10.648 - 11 + 1 = 0.648$$

$$f'(x_2) = 3(2.2)^2 - 5 = 3(4.84) - 5 = 14.52 - 5 = 9.52$$

$$x_3 = 2.2 - \frac{0.648}{9.52} = 2.2 - 0.068 = 2.132$$

$$f(x_3) = (2.132)^3 - 5(2.132) + 1 = 9.694 - 10.66 + 1 = 0.034$$

$$f'(x_3) = 3(2.132)^2 - 5 = 3(4.545) - 5 = 13.635 - 5 = 8.635$$

$$x_4 = 2.132 - \frac{0.034}{8.635} = 2.132 - 0.0039 = 2.128$$

CUARTO PUNTO

$\boldsymbol{a)e^x + 2^{-x} + 2coscosx - 6 = 0}$

$$f(x) = e^x + 2^{-x} + 2coscosx - 6 = 0$$
$$f(1) = e^1 + 2^{-1} + 2coscos(1) - 6 = 0.7152$$
$$f(2) = e^2 + 2^{-2} + 2coscos(2) - 6 = 3.6378$$
$$x_2 = 2 - \frac{3.6378(2-1)}{3.6378 - 0.7152} = 2 - \frac{3.6378}{2.9226} = 2 - 1.244 = 0.756$$
$$f(0.756) = -0.153$$
$$f(2) = 3.6378$$

$$x_3 = 0.756 - \frac{-0.153(0.756-2)}{-0.153 - 3.6378} = 0.756 - \frac{-0.153(-1.244)}{-3.7908} = 0.756 - 0.0503 = 0.706$$

$\boldsymbol{b)\ xlogx - 10 = 0}$

$f(5) = 5\log(5) - 10 = 5 - 0.69897 - 10 = 3.49485 - 10 = -6.50515$

$f(10) = 10\log(10) - 10 = 10 * 1 - 10 = 0$

$\boldsymbol{c)e^x + x^3 + 2x^2 + 10x - 20 = 0}$

$f(1) = e^1 + 1^3 + 2(1)^2 + 10(1) - 20 = 2.718 + 1 + 2 + 10 - 20 = -4.282$

$f(2) = e^2 + 2^3 + 2(2)^2 + 10(2) - 20 = 7.389 + 8 + 8 + 20 - 20 = 23.389$

$$x_2 = 2 - \frac{23.389(2-1)}{23.389 - (-4.282)} = 2 - \frac{23.389}{27.671} = 2 - 0.8454 = 1.1546$$

$$f(1) = e^{1.1546} + (1.1546)^3 + 2(1.1546)^2 + 10(1.1546) - 20$$
$$= 3.172 + 1.538 + 2.666 + 11.546 - 20 = -1.078$$
$$f(2) = 23.389$$

$$x_3 = 1.1546 - \frac{-1.078(1.1546-2)}{-1.078 - 23.389} = 1.1546 - \frac{-1.078(-0.8454)}{-24.467} = 1.1546 + 000.73$$
$$= 1.1919$$

$$f(1.1919) = e^{1.1919} + (1.1919)^3 + 2(1.1919)^2 + 10(1.1919) - 20 = -0.043$$
$$f(1.1546) = -1.078$$

$\boldsymbol{d)\ senx - \csc\csc x + 1 = 0}$

$$f(1) = \sin(1) - \frac{1}{\sin(1)} + 1 = 0.8415 - 1.187 + 1 = 0.6545$$

$$f(2) = \sin(2) - \frac{1}{\sin(2)} + 1 = 0.9093 - 1.1 + 1 = 0.8093$$

$$x_2 = 2 - \frac{0.8093(2-1)}{0.8093 - 0.6545} = 2 - \frac{0.8093}{0.1548} = 2 - 5.229 = -3.229$$

$$f(-3.229) = \sin(-3.229) - \frac{1}{\sin(-3.229)} + 1 = 0.0684 - 14.63 + 1 = -13.698$$